EXPERIMENT 2:

ARC SPECTRUM

Date of Experiment : 31-07-2026 & 03-08-2026
Date of Submission : 07-08-2026

Akshara SomalaRaju
IMS24017

AIM:

To photograph the arc spectrum of Copper and Brass and to identify the specimen.

THEORY:

An arc spectrum is an atomic line emission spectrum produced when an electric discharge passes across a narrow gap between two metal tips or graphite electrodes containing the sample. When a high potential difference is applied, intense thermal heating vaporises the metal atoms into a gaseous state within the arc gap. Collisions with high-energy electrons in the discharge excite the outer valence electrons of these vaporised neutral atoms to higher energy levels. As these excited electrons decay back down to lower energy states, they emit photons of discrete energies corresponding to the difference between the two energy levels. Because every element possesses a unique set of quantised atomic energy levels, the emitted light consists of a characteristic set of sharp spectral lines that act as an optical fingerprint for that element.

When light emitted by an arc source passes through a spectrograph or prism spectrometer, it is dispersed into its constituent wavelengths and recorded on a scale or photographic plate. Because the index of refraction of optical materials varies non-linearly with wavelength, the position recorded on a comparator scale or microscope does not change linearly with wavelength. To model this non-linear dispersion accurately over a targeted spectral region, Hartmann introduced an empirical dispersion relation that maps the linear scale reading directly to the wavelength of light. According to Hartmann's dispersion formula, the wavelength λ of a spectral line is inversely proportional to the difference between its measured position d and a reference position parameter d_0, offset by a baseline wavelength parameter λ_0.

$$\lambda(d) = \lambda_0 + \frac{C}{d - d_0}$$

To evaluate the three constants (λ_0, C, d_0), a reference mercury spectrum is recorded alongside the unknown spectrum. Three distinct reference lines with known wavelengths ($\lambda_1, \lambda_2, \lambda_3$) are selected, and their scale positions (d_1, d_2, d_3) are measured. Substituting these pairs into Hartmann's equation yields three simultaneous equations. Subtracting them pairwise eliminates λ_0, allowing d_0 to be isolated from the ratio of wavelength and scale differences. Once d_0 is known, the constants C and λ_0 are directly calculated to complete the spectrograph calibration.

$$A = \frac{\lambda_2 - \lambda_1}{\lambda_3 - \lambda_1} \qquad B = \frac{d_2 - d_1}{d_3 - d_1}$$

Hartmann's constant's equations are as follows:

$$\lambda_0 = \frac{A \cdot \lambda_3 - B \cdot \lambda_2}{A - B}$$

$$d_0 = \frac{B \cdot d_3 - A \cdot d_2}{B - A}$$

$$C = \frac{(\lambda_2 - \lambda_1)(d_0 - d_1)(d_0 - d_2)}{d_2 - d_1}$$

With the constants established, the system is calibrated for that wavelength range. To identify an line in the unknown metal spectrum, its scale reading d_x is measured under identical alignment conditions. Substituting d_x back into Hartmann's formula yields the exact wavelength λ_x.

PROCEDURE:

(i)Calibration using Hg lamp:

1. Place the spectrometer on a rigid table and connect the CCD detector to the PC via the provided USB cable. Open the *HOLMARC Spectra Analyser* software and log in.
2. Switch on the mercury lamp and position it directly in front of the spectrometer's slit. Click the Run icon in the software to start the live spectrum view.
3. Fine-tune the micrometer screw on the entrance slit to sharpen the spectrum, and adjust the Exposure Time under Spectrometer Control to ensure the signal is clear and not "Over Saturated".
4. Click Stop, then click the Acquire Single Spectrum icon to capture a stable trace. Save the spectrum data to your PC.
5. Zoom into the graph by right-clicking and dragging a box over the key intensity peaks to identify the exact pixel values (d_1, d_2, d_3) corresponding to the known mercury emission lines.
6. Use the measured pixel values (d_1, d_2, d_3) and their corresponding known wavelengths ($\lambda_1, \lambda_2, \lambda_3$) to solve for the Hartmann constants (λ_0, C, d_0).

(ii)Spectrum for copper rods:

1. Sharpen the copper rods and mount the copper rods in the spark gap or arc light source assembly in front of the spectrometer slit.
2. Safely turn on the high-voltage excitation source to produce the copper arc or spark, ensuring the light is aligned with the entrance slit.

3. In the software, click Run to observe the real-time spectrum. Adjust exposure time to get clear, sharp intensity peaks without saturation, then click Stop and Acquire Single Spectrum.
4. Zoom into the prominent spectral peaks and record their corresponding pixel positions (d).
5. Apply the Hartmann formula using the constants calculated from the mercury calibration.
6. Compare the calculated wavelengths (λ) against standard atomic emission wavelengths for copper.

(ii)Spectrum for brass rods:

1. Secure the sharpened brass rods in the arc/spark stand, aligning the gap directly with the entrance slit of the spectrometer.
2. Safely energise the excitation source to generate a stable arc or spark between the brass electrodes.
3. Click Run in the software to view the real-time intensity curve.
4. Adjust the exposure time so the strongest peaks are distinct without saturating the CCD detector.
5. Click Stop, then Acquire Single Spectrum and save your data file.
6. Zoom into the recorded graph to locate the pixel positions (d) corresponding to the major emission peaks.
7. Convert each pixel position d to its wavelength $\lambda(d)$ using the Hartmann calibration constants derived from the calibration.

OBSERVATIONS:

Table 1:

Calibrating spectrometer with low pressure Hg:

Colour of Spectral lines	Known wavelength (nm)	Corresponding pixel values(x-axis)
Violet	404.656	183
Blue	λ_1 = 435.833	d_1 = 940
Green	λ_2 = 546.096	d_2 = 2340
Yellow	λ_3 = 579.066	d_3 = 2565

Spectrum observed:

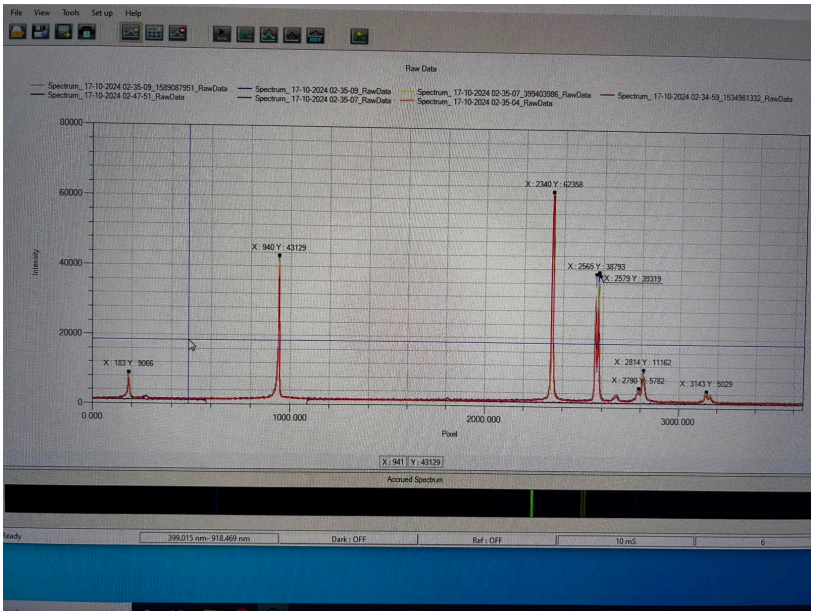


Table 2:

Wavelength of prominent lines in emission spectra of copper:

Distance of line from zero d_1(Pixel values)	Calculated wavelength $\lambda(d) = \lambda_0 + \frac{C}{d - d_0}$ (nm)	Theoretical Wavelength	Error Percentage (%)
1269	453.029	453.9	0.19
1297	454.659	458.7	0.88
1417	461.981	465.112	0.67
1993	507.07	510.554	0.68
2077	515.6	515.324	0.05
2293	539.8	531.641	1.53
2469	564.084	570.024	1.04
2501	568.914	578.213	1.61

Spectrum observed:

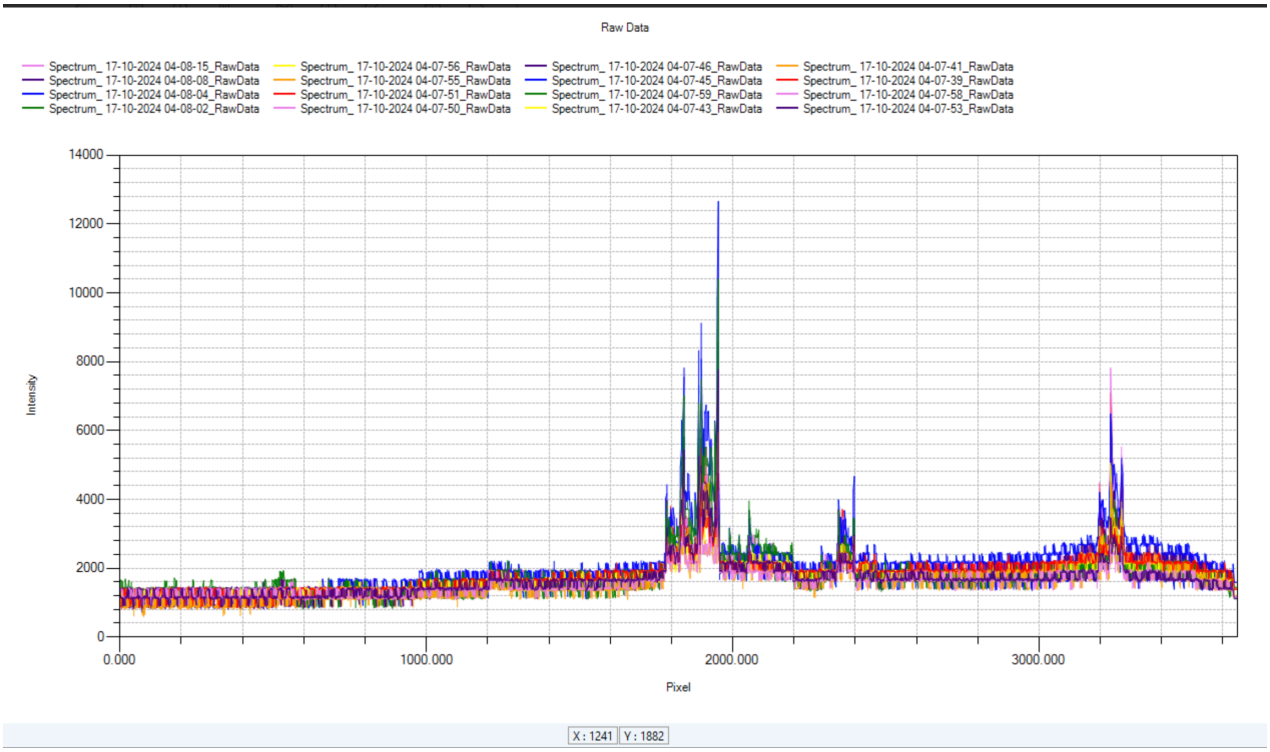
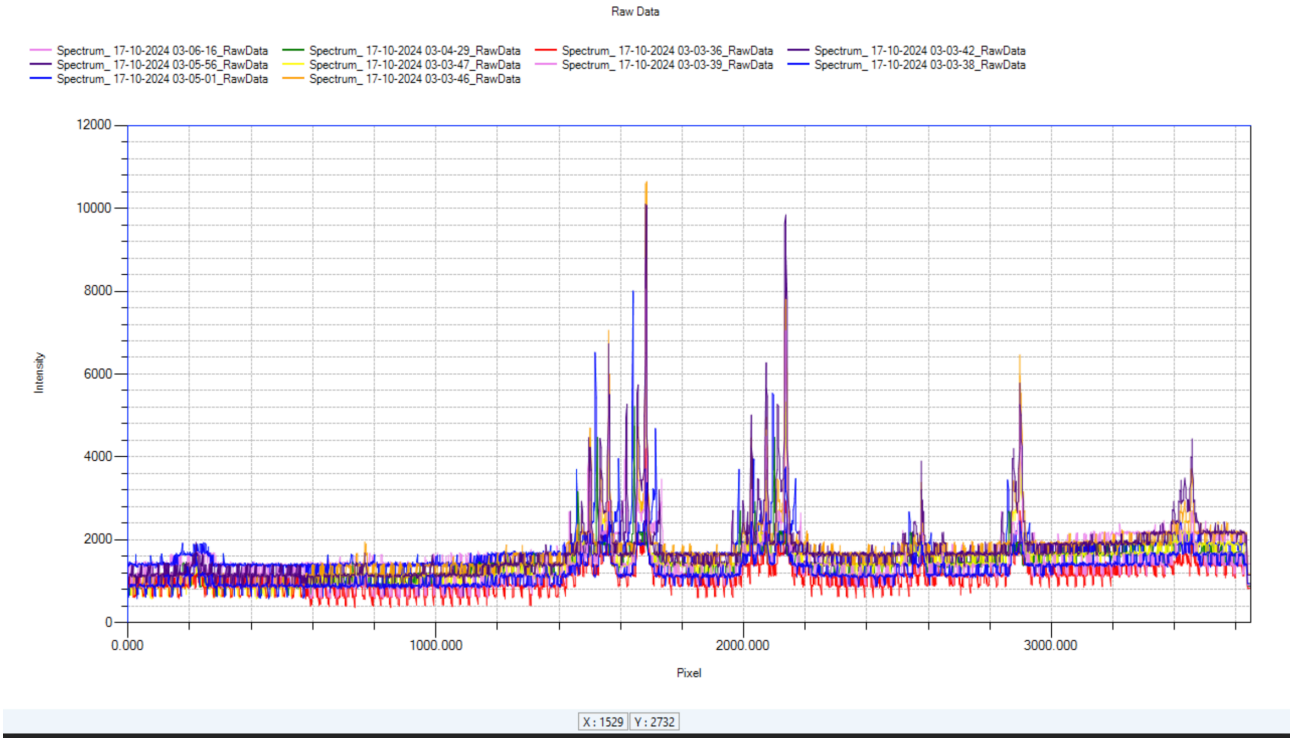


Table 3:

Wavelength of prominent lines in emission spectra of brass:

Distance of line from zero d_1 (Pixel values)	Calculated wavelength $\lambda(d) = \lambda_0 + \frac{C}{d - d_0}$ (nm)	Theoretical Wavelength	Element	Error Percentage (%)
1501	467.49	465.112	Cu	0.51
1565	471.8	472.01	Zn	0.04
1685	480.6	481.053	Zn	0.09
2073	515.2	515.324	Cu	0.02
2101	518.1	518.199	Zn	0.02
2137	521.9	521.641	Cu	0.05
2877	639.9	633.893	Zn	0.95

Spectrum observed:



CALCULATIONS AND ANALYSIS:

Calibration inputs:

$\lambda_1 = 435.833$ nm, $d_1 = 940$ px

$\lambda_2 = 546.096$ nm, $d_2 = 2340$ px

$\lambda_3 = 579.066$ nm, $d_3 = 2565$ px

Intermediate values:

$$A = \frac{\lambda_2 - \lambda_1}{\lambda_3 - \lambda_1} = \frac{546.096 - 435.833}{579.066 - 435.833} \approx 0.769816$$

$$B = \frac{d_2 - d_1}{d_3 - d_1} = \frac{2340 - 940}{2565 - 940} \approx 0.861538$$

Constants:

$$\lambda_0 = \frac{A\lambda_3 - B\lambda_2}{A - B} \approx 269.41 \text{ nm}$$

$$d_0 = \frac{B d_3 - A d_2}{B - A} \approx 4453.40 \text{ px}$$

$$C = \frac{(\lambda_2 - \lambda_1)(d_0 - d_1)(d_0 - d_2)}{d_2 - d_1} \approx 584813.2 \text{ nm} \cdot \text{px}$$

General formulas:

$$\lambda(d) = \lambda_0 + \frac{C}{d_0 - d} = 269.41 + \frac{584813.2}{4453.40 - d}$$

$$\text{Error (\%)} = \left| \frac{\lambda_{\text{exp}} - \lambda_{\text{theo}}}{\lambda_{\text{theo}}} \right| \times 100$$

For **Copper**:

-> $d = 1269$

$$\lambda(1269) = 269.41 + \frac{584813.2}{4453.40 - 1269} = 453.029 \text{ nm}$$

$$\text{Error} = \left| \frac{453.029 - 453.9}{453.9} \right| \times 100 = 0.19\,\%$$

-> $d = 1297$

$$\lambda(1297) = 269.41 + \frac{584813.2}{4453.40 - 1297} = 454.659 \text{ nm}$$

$$\text{Error} = \left| \frac{454.659 - 458.7}{458.7} \right| \times 100 = 0.88\,\%$$

-> $d = 1417$

$$\lambda(1417) = 269.41 + \frac{584813.2}{4453.40 - 1417} = 461.981 \text{ nm}$$

$$\text{Error} = \left| \frac{461.981 - 465.112}{465.112} \right| \times 100 = 0.67\,\%$$

-> $d = 1993$

$$\lambda(1993) = 269.41 + \frac{584813.2}{4453.40 - 1993} = 507.070 \text{ nm}$$

$$\text{Error} = \left| \frac{507.070 - 510.554}{510.554} \right| \times 100 = 0.68\,\%$$

-> $d = 2077$

$$\lambda(2077) = 269.41 + \frac{584813.2}{4453.40 - 2077} = 515.600 \text{ nm}$$

$$\text{Error} = \left| \frac{515.600 - 515.324}{515.324} \right| \times 100 = 0.05\,\%$$

-> $d = 2293$

$$\lambda(2293) = 269.41 + \frac{584813.2}{4453.40 - 2293} = 539.800 \text{ nm}$$

$$\text{Error} = \left| \frac{539.800 - 531.641}{531.641} \right| \times 100 = 1.53\,\%$$

-> $d = 2469$

$$\lambda(2469) = 269.41 + \frac{584813.2}{4453.40 - 2469} = 564.084 \text{ nm}$$

$$\text{Error} = \left| \frac{564.084 - 570.024}{570.024} \right| \times 100 = 1.04\,\%$$

-> $d = 2501$

$$\lambda(2501) = 269.41 + \frac{584813.2}{4453.40 - 2501} = 568.914 \text{ nm}$$

$$\text{Error} = \left| \frac{568.914 - 578.213}{578.213} \right| \times 100 = 1.61\,\%$$

For **Brass**:

$\text{-> } d = 1501(Cu)$

$$\lambda(1501) = 269.41 + \frac{584813.2}{4453.40 - 1501} = 467.490\ \text{nm}$$

$$\text{Error} = \left| \frac{467.490 - 465.112}{465.112} \right| \times 100 = 0.51\,\%$$

$\text{-> } d = 1565(Zn)$

$$\lambda(1565) = 269.41 + \frac{584813.2}{4453.40 - 1565} = 471.800\ \text{nm}$$

$$\text{Error} = \left| \frac{471.800 - 472.010}{472.010} \right| \times 100 = 0.04\,\%$$

$\text{-> } d = 1685(Zn)$

$$\lambda(1685) = 269.41 + \frac{584813.2}{4453.40 - 1685} = 480.600\ \text{nm}$$

$$\text{Error} = \left| \frac{480.600 - 481.053}{481.053} \right| \times 100 = 0.09\,\%$$

$\text{-> } d = 2073(Cu)$

$$\lambda(2073) = 269.41 + \frac{584813.2}{4453.40 - 2073} = 515.200\ \text{nm}$$

$$\text{Error} = \left| \frac{515.200 - 515.324}{515.324} \right| \times 100 = 0.02\,\%$$

-> $d = 2101(Zn)$

$$\lambda(2101) = 269.41 + \frac{584813.2}{4453.40 - 2101} = 518.100 \text{ nm}$$

$$\text{Error} = \left| \frac{518.100 - 518.199}{518.199} \right| \times 100 = 0.02\,\%$$

-> $d = 2137(Cu)$

$$\lambda(2137) = 269.41 + \frac{584813.2}{4453.40 - 2137} = 521.900 \text{ nm}$$

$$\text{Error} = \left| \frac{521.900 - 521.641}{521.641} \right| \times 100 = 0.05\,\%$$

-> $d = 2877(Zn)$

$$\lambda(2877) = 269.41 + \frac{584813.2}{4453.40 - 2877} = 639.900 \text{ nm}$$

$$\text{Error} = \left| \frac{639.900 - 633.893}{633.893} \right| \times 100 = 0.95\,\%$$

RESULT:

- The emission spectra of pure copper rods and brass rods were successfully recorded using a CCD-based spectrometer calibrated with a Mercury discharge lamp.
- The measured wavelength values for the pure copper arc closely matched the characteristic Cu I emission lines in the visible spectrum.
- The spectrum of the brass sample exhibited characteristic lines corresponding to both Copper (Cu) and Zinc (Zn), confirming the elemental alloy composition of brass.

DISCUSSION AND REMARKS:

The CCD-based spectrometer provides an efficient and accurate method for atomic emission spectroscopy, eliminating the subjective errors typical of manual visual alignment. By using a mercury discharge lamp as a standard

reference, the non-linear dispersion of the optical system was successfully mapped to the CCD detector pixels using Hartmann's dispersion relationship. The calculated calibration constants allowed for the conversion of raw pixel data into absolute wavelengths across the visible spectrum, demonstrating how critical precise initial calibration is to preventing systematic wavelength errors in subsequent measurements.

For the pure copper electrode arc, the observed emission peaks aligned closely with the characteristic atomic copper lines, particularly the distinct green feature in the five-hundred-ten to five-hundred-twenty nanometer region. When analysing the brass electrodes, the recorded spectrum clearly displayed a superposition of emission lines originating from both copper and zinc. The unambiguous appearance of prominent zinc atomic lines alongside the copper signature confirmed the binary alloy nature of the brass material.

Minor discrepancies between the experimental results and standard reference values can be attributed to factors such as thermal fluctuations in the arc gap, mechanical drift, and the finite spatial resolution of the CCD array. Additionally, local peak broadening caused by occasional over-saturation of high-intensity lines can introduce slight uncertainties in pinning down the exact peak pixel centres. These uncertainties can be minimised in future trials by maintaining a stabilised arc, optimising the entrance slit width, and carefully tuning detector exposure times to ensure signals remain within the linear response range.

LAB RECORD:

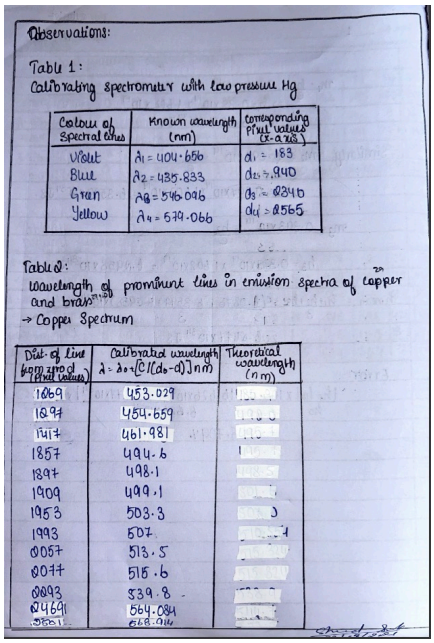
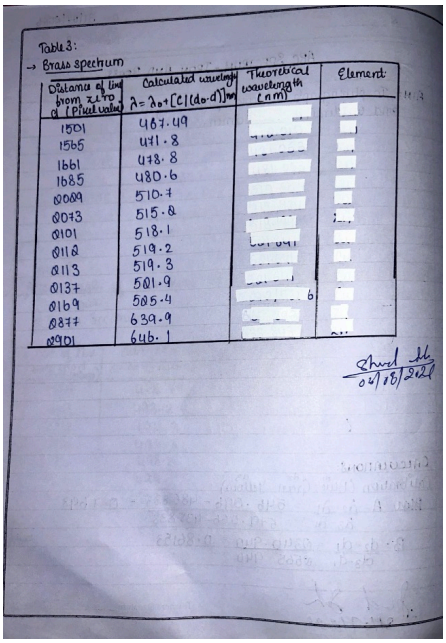

Observations:

Table 1:
Calibrating spectrometer with low pressure Hg

Colour of Spectral line	Known wavelength (nm)	corresponding Pixel values (x-axis)
Violet	$\lambda_1 = 404.656$	$d_1 = 183$
Blue	$\lambda_2 = 435.833$	$d_2 = 940$
Green	$\lambda_3 = 546.096$	$d_3 = 0340$
Yellow	$\lambda_4 = 579.066$	$d_4 = 0565$

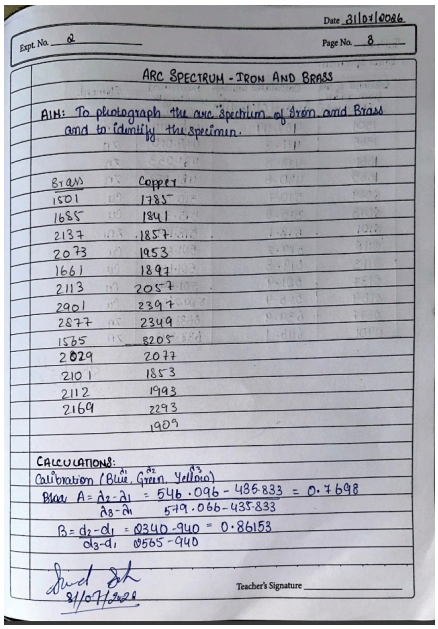
Table 2:
Wavelength of prominent lines in emission spectra of copper and brass
→ Copper Spectrum

Dist. of Line from zero d (Pixel value)	Calibrated wavelength $\lambda = \lambda_0 + [C/(d_0 - d)]$ nm	Theoretical wavelength (nm)
1869	453.029	
1897	454.659	
1917	461.981	
1857	494.6	
1897	498.1	
1909	499.1	
1953	503.3	
1993	507	
0057	513.5	
0077	515.6	
0093	539.8	
04691	564.084	
0501	568.916	


Table 3:
→ Brass spectrum

Distance of line from zero d (Pixel value)	Calculated wavelength $\lambda = \lambda_0 + [C/(d_0 - d)]$ nm	Theoretical wavelength (nm)	Element
1501	467.49		
1565	471.8		
1661	478.8		
1685	480.6		
0089	510.7		
0073	515.8		
0101	518.1		
0118	519.2		
0113	519.3		
0137	501.9		
0169	505.4		
0877	639.9		
0901	646.1		

04/08/2024


Date 31/01/2026
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ARC SPECTRUM - IRON AND BRASS

AIM: To photograph the arc spectrum of Iron and Brass and to identify the specimen.

Brass	Copper
1501	1785
1685	1841
2137	1857
2073	1953
1661	1897
2113	2057
2901	2397
2877	2349
1585	8205
2029	2077
2101	1853
2112	1993
2169	2293
	1909

CALCULATIONS:
Calibration (Blue, Green, Yellow)
Blue $A = \frac{\lambda_2 - \lambda_1}{\lambda_3 - \lambda_1} = \frac{546.096 - 435.833}{579.066 - 435.833} = 0.7698$
$B = \frac{d_2 - d_1}{d_3 - d_1} = \frac{0340 - 940}{0565 - 940} = 0.86153$

31/01/2026

Teacher's Signature



REFERENCES:

- https://www.scribd.com/doc/262144591/Arc-Spectrum
- CCD Based Constant Deviation Spectrometer (For Arc emission and Absorption Measurements) - Instruction Manual